

Product Requirements Document

Admin Panel Modules

1. Dashboard

Display overall business metrics.

Widgets

- Total Sales
- Total Service Calls
- Today's Jobs
- Pending Services
- Active AMC Customers
- Monthly Revenue
- Low Stock Alerts
- Employee Attendance
- Recent Invoice
- Customer Complaints
- Service Status Chart

2. Customer Management

Store customer information.

Fields:

- Customer ID
- Name
- Mobile
- Alternate Number
- Email
- Address
- GST Number (optional)
- Customer Type
 - Retail
 - Commercial

Features

- Customer History
- Purchase History
- Service History
- AMC History

3. Product Management

Products

Example

- AC
- Refrigerator
- Washing Machine
- Water Cooler
- Spare Parts

Fields

- Product Name
- Brand
- Model
- Serial Number
- Category
- Selling Price
- Purchase Price
- Warranty
- Stock Quantity

4. Quotation Management

Create quotations before sale.

Quotation Number

Customer

Products

Discount

GST

Total

Status

- Draft
- Sent
- Approved
- Rejected

Button

Convert to Proforma Invoice

5. Invoice Management

Generate GST Invoice

Features

- Invoice Number (Auto)
- Customer Details
- Product List
- Tax Calculation
- Discount
- Payment Method
- Print Invoice
- PDF Download
- WhatsApp Invoice
- Email Invoice

Status

- Paid
- Pending
- Partial Payment

6. Inventory Management

Stock Management

Features

- Purchase Entry
- Sales Entry
- Stock In
- Stock Out
- Warehouse
- Supplier

Stock Alert

- Low Stock
- Out of Stock
- Minimum Quantity Alert

Reports

- Current Stock
- Stock Value
- Fast Moving Products
- Slow Moving Products

7. Supplier Management

- Supplier Details
- Purchase History
- Pending Payments
- GST Details

8. Purchase Module

Purchase Invoice

- Supplier
- Products
- Quantity
- Cost
- GST
- Total

Updates inventory automatically.

9. Service Management

Every service request becomes a Job Card.

Fields

- Job Number
- Customer
- Product
- Complaint
- Technician
- Service Date
- Priority

Status

- Pending
- Assigned
- In Progress
- Completed

- Cancelled

10. Service Tracker

Track every service.

Timeline

Complaint Received

↓

Technician Assigned

↓

Visited

↓

Repair Done

↓

Payment Collected

↓

Completed

11. AMC Tracker

Annual Maintenance Contract

Store

- AMC Number
- Customer
- Product
- Start Date
- End Date
- Number of Visits
- Remaining Visits

Alerts

- AMC Expiry
- Upcoming Visit
- Missed Visit

12. Installation Tracker

After Sales

Status

- Product Delivered
- Installation Pending
- Technician Assigned
- Installed
- Customer Verified

13. Complaint Tracker

Complaint Number

Customer

Issue

Priority

Technician

Status

- Open

- In Progress
- Closed

14. Status Tracker

A common status tracker for all jobs.

Filters

- New
- Assigned
- Pending Parts
- On Hold
- Completed
- Cancelled

15. Employee Management

Employee Details

- Employee ID
- Name
- Phone
- Address
- Designation
- Salary
- Joining Date
- Documents

Roles

- Admin

16. Attendance

Daily Attendance

- Check In
- Check Out
- Working Hours
- Late Entry
- Leave
- Leave Reason

Reports

- Daily
- Weekly
- Monthly

17. Technician Module

Assigned Jobs

Today's Jobs

Completed Jobs

Location

Service History

Parts Used

18. Payment Management

Payments

- Cash
- UPI
- Card
- Bank

Pending Payments
Payment Reminder
Receipt Generation

19. Reports

Sales Report
Service Report
Inventory Report
AMC Report
Employee Attendance Report
Customer Report
Purchase Report
Profit & Loss
GST Report

20. Notifications

SMS
WhatsApp
Email
Low Stock Alert
AMC Expiry
Service Reminder
Payment Reminder

21. User & Role Management

Permissions

- Admin

Suggested Database Structure

Customers
Employees
Products
Categories
Suppliers
Purchases
Purchase Items
Quotations
Quotation Items
Invoice
Invoice Items
Inventory
Stock Movements
Services
Job Cards

AMC Contracts
Complaints
Attendance
Payments
Notifications
Users
Roles
Permissions

Dashboard Layout

Dashboard

Revenue Service Jobs Pending AMC Stock Alert

Today's Jobs

Pending Services

Completed Services

Sales Graph Service Graph

Low Stock

Recent Invoice

Technician Status

Attendance Summary

Recommended Technology Stack

Frontend

- React.js (or Next.js)
- Tailwind CSS
- Material UI / Shadcn UI

Backend

- Node.js + Express
- REST API (or NestJS for larger projects)

Database

- MySQL or PostgreSQL

Authentication

- JWT
- Role-Based Access Control (RBAC)

Reports & Documents

- PDF invoice generation
- Excel export

Storage

- Cloudinary or AWS S3 for images/documents

Future Enhancements

- Customer portal to view invoice and AMC details
- Mobile app for technicians
- GPS tracking of service visits
- Barcode/QR code support for inventory
- WhatsApp integration for quotations, invoice, and reminders
- Online payment gateway integration
- Business analytics with sales and service forecasting
- Multi-branch support
- Backup and audit logs

Application Flow, Navigation & User Journey

1. Overall Application Flow

Login



Dashboard



- └─ Stock Out
- └─ Warehouses
- └─ Stock Alerts
- └─ Suppliers
- └─ Supplier List
- └─ Purchase History
- └─ Payments
- └─ Services
- └─ New Complaint
- └─ Job Cards
- └─ Service Tracker
- └─ Installation
- └─ AMC Tracker
- └─ Employees
- └─ Employee List
- └─ Attendance
- └─ Leave
- └─ Technician Jobs
- └─ Reports
- └─ Notifications
- └─ Settings

2. Left Sidebar Navigation

 Dashboard

 Customers
Customer List
Add Customer

 Sales
Quotations
Invoices
Payments

 Products
Products
Categories
Brands

 Inventory
Purchase
Stock In
Stock Out
Stock Alert
Warehouses

 Suppliers

 Service Management

Complaints
Job Cards
Service Tracker
Installation
AMC Tracker
Status Tracker

 Employees
Employees
Attendance
Leave
Technician Jobs

 Reports

 Notifications

 Settings
User Roles
Company
Tax
Backup

3. Dashboard Navigation

Dashboard

Revenue Card
Total Sales



Click

Sales Module

Today's Service



Job Card

Low Stock



Inventory

AMC Expiring



AMC Module

Attendance



Employee Attendance

Complaint



Complaint Details

4. Customer Journey

Customer Calls



Create Customer



Customer Profile



Create Quotation



Quotation Approved



Convert to Invoice



Payment



Product Delivery



Installation



Warranty Starts



AMC Offer



Service Requests



Customer History

5. Sales Journey

New Customer



Quotation



Quotation Approval



Convert to Invoice



Receive Payment



Inventory Reduced



Generate GST Invoice



Send Invoice



Delivery



Installation

6. Service Journey

Customer Complaint



Create Complaint



Generate Job Card



Assign Technician



Technician Visit



Parts Required?

Yes



Inventory Check



Issue Parts



Repair

No



Repair



Service Completed



Collect Payment



Close Job



Customer Feedback

7. AMC Journey

Customer Purchase



Warranty Ends



AMC Created



AMC Active



Visit Reminder



Technician Visit



Service Completed



Remaining Visits Updated



AMC Renewal Reminder

8. Inventory Journey

Supplier



Purchase Entry



Stock In



Sales Invoice



Stock Out



Service Parts Used



Stock Updated



Minimum Stock



Alert Generated



Purchase Order

9. Technician Journey

Today's Jobs



Accept Job



Navigate to Customer



Start Job



Update Status



Use Spare Parts



Upload Photos



Collect Signature



Complete Job



Payment Status



Next Job

10. Invoice Journey

Quotation



Approved



Generate Invoice



GST Calculation



Payment



Print



PDF



WhatsApp



Email



Reports

11. Purchase Flow

Supplier



Purchase Order



Receive Products



Purchase Invoice



Stock Updated



Supplier Payment



Purchase Report

12. Complaint Flow

Complaint Received



Register Complaint



Assign Technician



Visit



Repair



Customer Approval



Complaint Closed

13. Admin Daily Workflow

Login



Dashboard



Check Today's Revenue



Review Pending Services



Approve Quotations



Check Low Stock



Create Purchase Order



Assign Technician



Review Payments



View Reports



Logout

15. Screen Navigation Hierarchy

Login



└─ Dashboard



└─ Customers



└─ List



└─ Add



└─ Profile



└─ Products



└─ List



└─ Categories



└─ Brands



└─ Sales



└─ Quotations



└─ New



└─ Edit



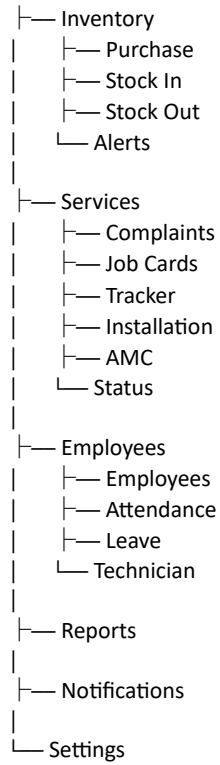
└─ Convert Invoice



└─ Invoices



└─ Payments



16. Role-Based Navigation

Module Admin

Dashboard 

Customers 

Products 

Quotations 

Invoices 

Inventory 

Suppliers 

Purchase 

Service Jobs 

AMC 

Installation 

Complaints 

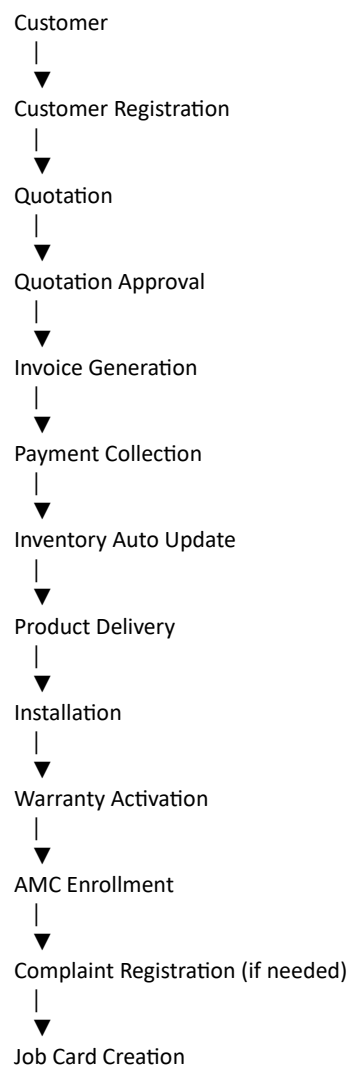
Attendance 

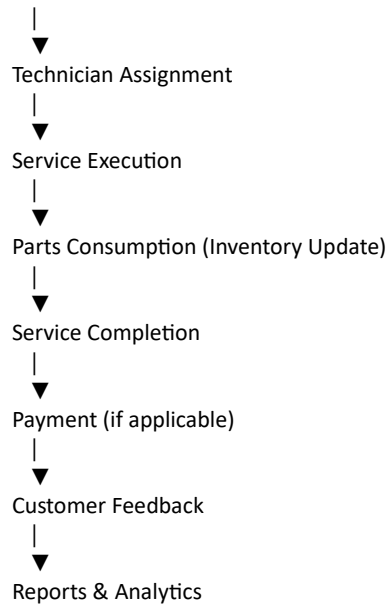
Employees 

Reports 

Settings 

Recommended End-to-End Business Workflow





UI/UX DESIGN BRIEF

Design Element	UI/UX Design Brief
Overall Aesthetic	Modern Enterprise ERP , clean, professional, minimal, dashboard-first. Inspired by Linear , Notion , Monday.com , and Zoho with spacious layouts, clear information hierarchy, and subtle visual depth. The interface should emphasize operational efficiency with quick scanning and low cognitive load.
Primary Color	#2F612F (Forest Green) – Primary brand color used for navigation, headers, primary buttons, active menu items, charts, and success indicators.
Secondary Color	#4D8A4D – A lighter green used for hover states, secondary buttons, progress indicators, badges, and charts.
Accent / CTA Color	#FFF101 (Bright Yellow) – Used sparingly for primary CTAs, notifications, pending tasks, highlights, stock alerts, and important actions. Never use as a full background.
Background Color	#F7F8F6 (Main Background) #FFFFFF (Cards & Modals) #EEF2EE (Sidebar Sections)
Surface Colors	Sidebar: #2F612F Cards: #FFFFFF Hover State: #F1F5F2 Selected Row: #E7F4E7
Text Color	Primary: #1E1E1E Secondary: #5E6470 Muted: #8B9098 White Text: #FFFFFF (on green backgrounds)
Status Colors	Success: #2F612F Warning: #FFF101 Error: #E53935 Info: #2196F3
Typography	Font: Inter (Primary UI Font) Fallback: SF Pro Display / Roboto
Heading Sizes	H1 – 32px / 700 H2 – 26px / 700 H3 – 22px / 600 Section Title – 18px / 600
Body Text	Body Large – 16px / 400 Body – 14px / 400 Small Labels – 12px / 500
Buttons	Primary: Green (#2F612F) with white text Secondary: White with green outline Warning: Yellow background with dark text
Border Radius	10px for cards, forms, buttons, tables, modals. Soft modern corners throughout.
Component Style	Flat design with subtle elevation . Minimal shadows (4–8px blur), clean borders (#E6E8EB), generous whitespace, and consistent spacing using an 8px grid system .

Design Element	UI/UX Design Brief
Cards	White cards, 10px radius, soft shadow, icon + metric + trend indicator. Cards should be interactive with hover elevation.
Tables	Sticky headers, zebra row hover, inline search, sortable columns, filters, pagination, row actions on hover, status chips.
Forms	Two-column desktop layout, single-column mobile. Floating labels, inline validation, searchable dropdowns, step-based forms for invoices and job cards.
Modals	Medium-width (600–800px), slide/fade animation, rounded corners, keyboard accessible, primary action fixed at bottom-right.
Sidebar	Fixed left sidebar (280px), dark green background (#2F612F), white icons, collapsible, grouped navigation with expandable menus.
Top Navigation	White top bar with global search, notifications, quick actions (+ Invoice, + Service), user profile, branch selector.
Dashboard Pattern	KPI cards → Charts → Recent Activity → Tables → Alerts. Every widget should be clickable for drill-down navigation.
Navigation Style	Desktop: Left Sidebar + Top Header Tablet: Collapsible Sidebar Mobile: Bottom Navigation + Hamburger Menu
Icons	Lucide Icons or Heroicons (2px stroke), consistent outlined style with filled states for active items.
Charts	Minimal line charts, bar charts, donut charts using green, yellow, blue, and gray. Avoid excessive gradients.
Shadows	Very subtle shadows only: 0 2px 8px rgba(0,0,0,0.06) for cards and dropdowns. No heavy neumorphism or glassmorphism.
Dark / Light Mode	Light Mode as default for office productivity. Optional Dark Mode with dark green navigation (#1B2D1B), dark gray surfaces (#1E1F24), and yellow highlights retained for CTAs.
Reference Apps	Linear, Notion, Stripe Dashboard, Zoho Books, Monday.com, Atlassian Jira, Vercel Dashboard, Odoo ERP
Micro-interactions	200ms transitions, button ripple/hover, loading skeletons, animated KPI counters, status badge transitions, success toasts, progress indicators for service workflows.
Key UI Patterns	Dashboard Cards • Data Tables • Stepper Forms • Side Drawers • Timeline (Service Tracker) • Status Chips • Kanban View (Jobs) • Calendar View (AMC Visits) • Searchable Dropdowns • Floating Action Buttons (Mobile)
Responsive Design	Desktop-first (1440px), Tablet (1024px), Mobile (390px+). Responsive grids, collapsible tables, touch-friendly controls (minimum 44px touch targets).
Accessibility	WCAG 2.1 AA compliant. Minimum 4.5:1 text contrast, keyboard navigation, visible focus states, scalable typography (minimum 14px), color-independent status indicators, screen reader-friendly labels, and ARIA support.
Spacing System	8px spacing scale (8, 16, 24, 32, 40, 48px). Maximum content width: 1440px. Card padding: 24px. Form spacing: 20px.
Animation Style	Smooth, subtle, purposeful. No distracting motion. Fade, slide, and scale animations limited to 150–250ms using ease-in-out curves.

Backend Schema (Data Model & Authentication)

Backend	Specification
Table: users	id (uuid PK), employee_id (FK → employees.id), role_id (FK → roles.id), full_name (text), email (unique), phone (text), password_hash (text), avatar_url (text), status (active/inactive), last_login (timestamp), created_at, updated_at
Table: roles	id (uuid PK), role_name (Admin, Manager, Sales Executive, Technician, Accountant), description, created_at
Table: permissions	id (uuid PK), module_name, action (view/create/update/delete/export/approve), description
Table: role_permissions	id, role_id (FK → roles.id), permission_id (FK → permissions.id)
Table: employees	id, employee_code, full_name, designation, department, phone, email, salary, joining_date, status, documents, created_at
Table: customers	id, customer_code, customer_type, company_name, customer_name, mobile, alternate_mobile, email, gst_number, address, city, state, pincode, created_by (FK → users.id), created_at
Table: products	id, sku, product_name, category_id (FK), brand_id (FK), model, serial_number, warranty_months, purchase_price, selling_price, tax_rate, stock_quantity, created_at
Table: categories	id, category_name, created_at
Table: brands	id, brand_name, created_at
Table: suppliers	id, supplier_code, company_name, contact_person, phone, email, gst_number, address, created_at
Table: purchase_orders	id, po_number, supplier_id (FK), purchase_date, subtotal, gst_amount, total_amount, payment_status, created_by (FK → users.id)
Table: purchase_items	id, purchase_order_id (FK), product_id (FK), quantity, unit_price, tax_amount, total
Table: inventory	id, product_id (FK), warehouse_id (FK), available_stock, reserved_stock, minimum_stock, reorder_level, updated_at
Table: stock_movements	id, product_id (FK), movement_type (IN/OUT), reference_type (Purchase/Sale/Service), reference_id, quantity, remarks, created_at

Backend	Specification
Table: quotations	id, quotation_number, customer_id (FK), quotation_date, expiry_date, subtotal, discount, gst_amount, grand_total, status (Draft/Sent/Approved/Rejected), created_by (FK)
Table: quotation_items	id, quotation_id (FK), product_id (FK), quantity, unit_price, discount, tax_amount, total
Table: invoices	id, invoice_number, quotation_id (FK), customer_id (FK), invoice_date, subtotal, discount, gst_amount, grand_total, payment_status, payment_method, created_by
Table: invoice_items	id, invoice_id (FK), product_id (FK), quantity, selling_price, tax_amount, total
Table: payments	id, invoice_id (FK), customer_id (FK), payment_method, payment_reference, amount, payment_date, remarks
Table: complaints	id, complaint_number, customer_id (FK), product_id (FK), complaint_description, priority, status, created_at
Table: job_cards	id, job_number, complaint_id (FK), technician_id (FK → employees.id), visit_date, status, estimated_cost, actual_cost, notes
Table: service_history	id, job_card_id (FK), technician_id (FK), work_done, spare_parts_used, customer_signature, completed_at
Table: service_status_logs	id, job_card_id (FK), previous_status, current_status, updated_by (FK → users.id), updated_at
Table: amc_contracts	id, amc_number, customer_id (FK), product_id (FK), start_date, end_date, total_visits, completed_visits, remaining_visits, status
Table: amc_visits	id, amc_id (FK), visit_number, scheduled_date, completed_date, technician_id (FK), remarks
Table: installations	id, invoice_id (FK), customer_id (FK), technician_id (FK), installation_date, installation_status, customer_confirmation
Table: attendance	id, employee_id (FK), attendance_date, check_in, check_out, working_hours, attendance_status, leave_reason
Table: leave_requests	id, employee_id (FK), leave_type, start_date, end_date, reason, approval_status, approved_by (FK)

Backend	Specification
Table: notifications	id, user_id (FK), title, message, notification_type (SMS/Email/WhatsApp), is_read, created_at
Table: audit_logs	id, user_id (FK), module_name, action, record_id, old_data (JSON), new_data (JSON), ip_address, created_at
Table: company_settings	id, company_name, gst_number, logo_url, invoice_prefix, financial_year, timezone, created_at
Relationships	users.role_id → roles.id (Many-to-One) users.employee_id → employees.id (One-to-One) customers.id → quotations.customer_id (One-to-Many) quotations.id → invoices.quotation_id (One-to-One) invoices.id → invoice_items.invoice_id (One-to-Many) suppliers.id → purchase_orders.supplier_id (One-to-Many) purchase_orders.id → purchase_items.purchase_order_id (One-to-Many) products.id → inventory.product_id (One-to-Many) customers.id → complaints.customer_id (One-to-Many) complaints.id → job_cards.complaint_id (One-to-One) employees.id → job_cards.technician_id (One-to-Many) job_cards.id → service_history.job_card_id (One-to-Many) customers.id → amc_contracts.customer_id (One-to-Many)
Auth Provider	JWT Authentication (Access Token + Refresh Token), Email/Password Login, Refresh Token Rotation, Password Reset (Email OTP), Optional Google OAuth, Optional Two-Factor Authentication (TOTP/Email OTP), Session Management with Redis
Row Level Security	Admin: Full access to all modules. Manager: Branch-level operational access. Sales Executive: Customers, Quotations, Invoices, Payments created by self/branch. Technician: Assigned Job Cards, Service History, AMC Visits, Attendance only. Accountant: Financial modules (Invoices, Payments, Purchases, Reports). Customer Portal: Access only to own Quotations, Invoices, AMC, and Service History.
User Roles	Admin – Full System Access Manager – Sales, Inventory, Service, Employees, Reports Sales Executive – Customers, Quotations, Invoices, Payments Technician – Assigned Jobs, AMC, Installations, Attendance Accountant – Purchases, Payments, GST, Financial Reports
File Storage	company/logo/employees/{employee_id}/documents/customers/{customer_id}/attachments/products/{product_id}/images/services/{job_id}/photos/quotations/{quotation_id}.pdf invoices/{invoice_id}.pdf receipts/{payment_id}.pdf Recommended: AWS S3 / Cloudinary / Supabase Storage
Sensitive Fields	password_hash → Argon2id encrypted refresh_token → Hashed salary → HR/Admin only payment_reference → Masked in UI bank_account_number → AES-256 encrypted upi_id → Encrypted smtp_password, jwt_secret, api_keys, whatsapp_token → Environment Variables / Secret Manager (never stored in DB)

Database Architecture

Item	Specification
Database Type	Relational Database Management System (RDBMS)
Recommended Database	PostgreSQL 16 (Recommended) / MySQL 8
ORM	Prisma ORM (Preferred) or TypeORM
Architecture	Normalized Database (3NF)
Primary Key Strategy	UUID (gen_random_uuid())
Foreign Keys	Enforced with Referential Integrity

Item	Specification
Indexes	UUID, Invoice Number, Customer Mobile, Product SKU, Job Number, AMC Number, Employee Code
Transactions	ACID Compliant Database Transactions
Soft Delete	deleted_at, deleted_by
Audit Trail	Every Update/Delete stored in Audit Logs
Migration Tool	Prisma Migrate

Relational Database Design

Users

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Roles

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Role Permissions

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Permissions

Customers

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Quotations Complaints

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Invoices Job Cards

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Payments Service History

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Installations Service Status Logs

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AMC Contracts

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AMC Visits

Products

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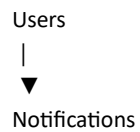
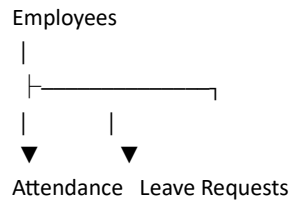
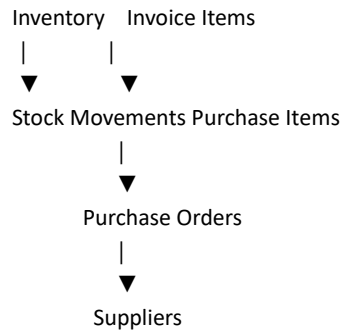
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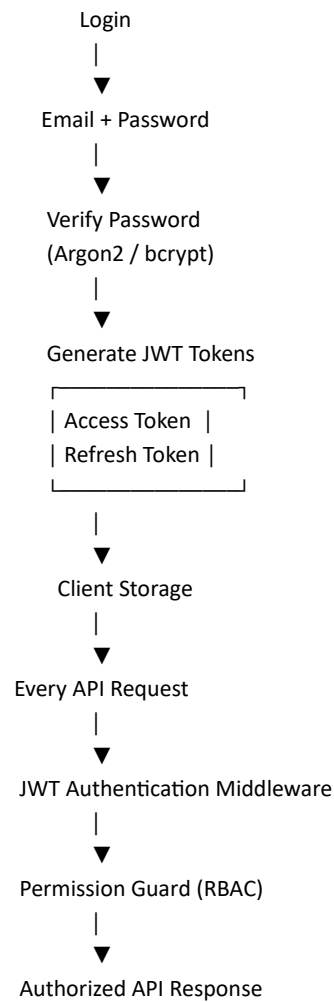


Entity Relationship Summary

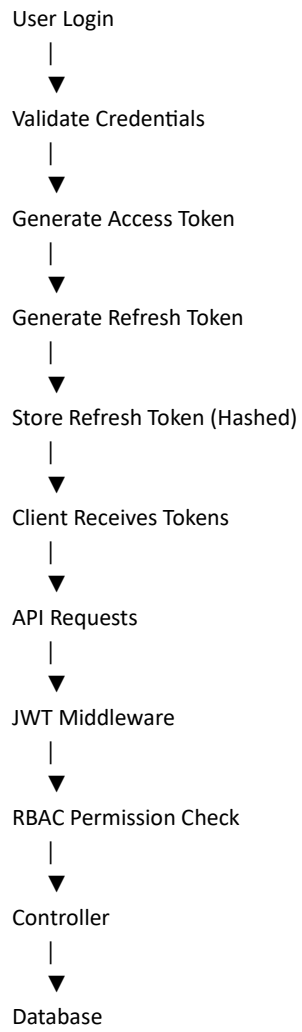
Parent Table	Child Table	Relationship
Roles	Users	One-to-Many
Roles	Role Permissions	One-to-Many
Permissions	Role Permissions	One-to-Many
Employees	Users	One-to-One
Customers	Quotations	One-to-Many
Customers	Invoices	One-to-Many
Customers	Complaints	One-to-Many
Customers	AMC Contracts	One-to-Many
Quotations	Quotation Items	One-to-Many
Quotations	Invoices	One-to-One
Invoices	Invoice Items	One-to-Many
Invoices	Payments	One-to-Many
Suppliers	Purchase Orders	One-to-Many
Purchase Orders	Purchase Items	One-to-Many
Products	Inventory	One-to-One

Parent Table	Child Table	Relationship
Products	Invoice Items	One-to-Many
Products	Purchase Items	One-to-Many
Products	Stock Movements	One-to-Many
Complaints	Job Cards	One-to-One
Job Cards	Service History	One-to-Many
Job Cards	Service Status Logs	One-to-Many
AMC Contracts	AMC Visits	One-to-Many
Employees	Attendance	One-to-Many
Employees	Leave Requests	One-to-Many
Users	Audit Logs	One-to-Many
Users	Notifications	One-to-Many

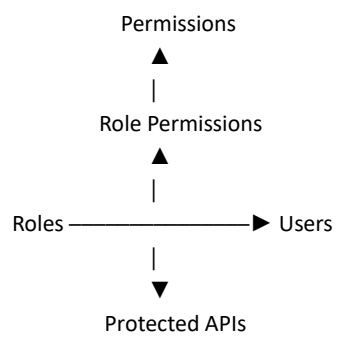
Authentication Architecture (JWT)



JWT Authentication Flow



Role-Based Access Control (RBAC)



User Role Hierarchy

Admin

Module Permission Matrix

Module	Admin
Dashboard	
Customers	CRUD
Products	CRUD
Inventory	CRUD
Quotations	CRUD
Invoices	CRUD
Payments	CRUD
Purchase	CRUD
Suppliers	CRUD
Complaints	CRUD
Job Cards	CRUD
AMC	CRUD
Installation	CRUD
Employees	CRUD
Attendance	CRUD
Reports	All
Settings	CRUD

Backend Folder Structure (NestJS)

```
src/  
├─ auth/  
├─ users/  
├─ roles/  
├─ permissions/  
├─ customers/  
├─ products/  
├─ inventory/  
├─ suppliers/  
├─ purchases/  
├─ quotations/  
├─ invoices/  
├─ payments/  
├─ complaints/  
└─ services/
```

```
├─ amc/
├─ installations/
├─ employees/
├─ attendance/
├─ notifications/
├─ reports/
├─ settings/
├─ common/
├─ prisma/
└─ main.ts
```

Recommended Production Stack

Layer	Technology
Framework	NestJS (TypeScript)
Database	PostgreSQL 16
ORM	Prisma ORM
Authentication	JWT (Access + Refresh Tokens)
Authorization	RBAC (Role-Based Access Control)
Password Hashing	Argon2id
Cache	Redis
Queue	BullMQ
API Documentation	Swagger (OpenAPI 3)
Storage	AWS S3 / Cloudinary
Logging	Winston + Morgan
Deployment	Docker + Nginx + PostgreSQL + Redis
Monitoring	Prometheus + Grafana + Sentry

Implementation Plan

Phase	Implementation Plan
Phase 1: Project Setup & Architecture	Initialize Git repository, configure monorepo (Frontend + Backend), set up NestJS, React/Next.js, PostgreSQL, Prisma ORM, Tailwind CSS, environment variables, Docker, ESLint, Prettier, Husky, CI/CD pipeline, logging, error handling, folder structure, API conventions, Swagger, and Git branching strategy.
Phase 2: Database & Backend Foundation	Design relational database schema, create Prisma models, generate migrations, seed master data (Roles, Permissions, Categories, Brands), configure UUIDs, indexes, audit tables, soft deletes, Redis, file storage, backup strategy, and repository layer.
Phase 3: Authentication & Authorization	Implement JWT Authentication (Access + Refresh Tokens), Login, Logout, Password Reset, Email Verification, RBAC (Role-Based Access Control), Permission Guards, Protected APIs, User Profile, Session Management, Activity Logs, and Security Middleware.

Phase	Implementation Plan
Phase 4: Dashboard Module	Develop dashboard APIs and UI with KPI cards, revenue summary, today's jobs, pending services, AMC overview, low stock alerts, employee attendance, recent invoices, complaint summary, charts, widgets, and global search.
Phase 5: User & Role Management	Build User CRUD, Employee-to-User mapping, Role Management, Permission Matrix, Branch Assignment (optional), Profile Settings, Password Management, Avatar Upload, and User Activity Logs.
Phase 6: Customer Management	Customer CRUD, Customer Profile, Contact Information, Address Management, Customer History, Purchase History, Service History, AMC History, Customer Search, Import/Export, and Duplicate Detection.
Phase 7: Product & Master Data	Product CRUD, Categories, Brands, Units, Product Images, Warranty Details, Pricing, Product Search, Barcode/QR support (optional), Product Status, and Master Data Management.
Phase 8: Supplier Management	Supplier CRUD, Supplier Contacts, GST Details, Supplier Ledger, Purchase History, Pending Payments, Supplier Reports, and Vendor Performance.
Phase 9: Purchase Management	Purchase Orders, Purchase Invoices, Purchase Items, Supplier Payments, Purchase Approval Workflow, Inventory Auto Update, Purchase Reports, and Purchase Returns.
Phase 10: Inventory Management	Warehouse Management, Stock In, Stock Out, Stock Transfers, Inventory Adjustments, Low Stock Alerts, Reorder Levels, Stock Valuation, Batch Tracking (optional), and Stock Movement Logs.
Phase 11: Quotation Management	Quotation CRUD, Product Selection, Discount Engine, GST Calculation, PDF Generation, Email/WhatsApp Sharing, Approval Workflow, Expiry Management, and Convert Quotation → Invoice.
Phase 12: Invoice & Billing	GST Invoice Generation, Invoice Items, Payment Terms, Payment Status, Invoice PDF, Print, Email, WhatsApp Sharing, Credit Notes (optional), Invoice Reports, and Auto Inventory Deduction.
Phase 13: Payment Management	Payment Collection, Cash/UPI/Card/Bank Support, Partial Payments, Outstanding Tracking, Receipts, Payment History, Payment Reminders, and Customer Ledger.
Phase 14: Complaint Management	Complaint Registration, Complaint Categories, Priority Assignment, Complaint Timeline, Status Updates, Customer Notifications, Complaint Reports, and SLA Tracking.
Phase 15: Service Management	Job Card Creation, Technician Assignment, Service Scheduling, Service Status Updates, Spare Parts Usage, Service Notes, Customer Signature, Photo Upload, Service Completion, and Service Billing.
Phase 16: Service Tracker	Real-time Job Status Timeline, Technician Progress, Customer Notifications, Delay Tracking, Pending Parts Workflow, Service Dashboard, and Status History Logs.
Phase 17: AMC Management	AMC Contract Creation, Visit Scheduling, Renewal Reminders, Visit History, AMC Dashboard, Remaining Visit Tracking, Expiry Notifications, and AMC Reports.
Phase 18: Installation Management	Installation Scheduling, Technician Assignment, Installation Checklist, Customer Confirmation, Installation Status Tracking, and Installation Reports.
Phase 19: Employee Management	Employee CRUD, Departments, Designations, Documents, Salary Information, Technician Profiles, Employment Status, and Employee Search.
Phase 20: Attendance & Leave	Daily Attendance, Check-In/Check-Out, Working Hours Calculation, Leave Requests, Leave Approval Workflow, Attendance Reports, Overtime Tracking, and Monthly Summary.
Phase 21: Technician Portal	Technician Dashboard, Assigned Jobs, Today's Schedule, Service Updates, Spare Parts Consumption, GPS Location (optional), Customer Signature, Offline Support (future), and Job Completion.
Phase 22: Notifications Module	Email Notifications, WhatsApp Integration, SMS Gateway, Low Stock Alerts, AMC Expiry Alerts, Service Reminders, Payment Reminders, Push Notifications (future), and Notification History.
Phase 23: Reports & Analytics	Sales Reports, Purchase Reports, Inventory Reports, Service Reports, AMC Reports, Attendance Reports, Financial Reports, GST Reports, Customer Reports, Technician Performance, and Export (PDF/Excel).

Phase	Implementation Plan
Phase 24: Company Settings	Company Profile, GST Configuration, Invoice Templates, Tax Settings, Notification Settings, Email Configuration, Backup Settings, Branch Management (optional), and Branding.
Phase 25: API Optimization	Pagination, Filtering, Sorting, Search APIs, Database Indexing, Query Optimization, Redis Caching, API Rate Limiting, Background Jobs, and Performance Profiling.
Phase 26: UI/UX Polish	Responsive Design, Skeleton Loaders, Empty States, Error Pages, Form Validation, Accessibility (WCAG AA), Keyboard Navigation, Dark Mode, Micro-interactions, and Animation Refinement.
Phase 27: Quality Assurance & Testing	Unit Testing, Integration Testing, API Testing, End-to-End Testing, Role-Based Testing, Cross-Browser Testing, Mobile Responsiveness, Performance Testing, Security Testing, and UAT.
Phase 28: Deployment & Production	Production Environment, Docker Deployment, Nginx Configuration, SSL, Domain Setup, CI/CD Pipeline, Database Backup, Monitoring, Logging, Health Checks, and Production Documentation.
Phase 29: Go-Live & Support	Data Migration, User Training, Production Monitoring, Bug Fixes, Performance Tuning, Feature Stabilization, Documentation Handover, and Maintenance Plan.

Recommended Development Timeline

Week 1–2

- ✓ Project Setup
- ✓ Database
- ✓ Authentication

Week 3

- ✓ Dashboard
- ✓ User & Roles

Week 4

- ✓ Customer Management
- ✓ Product Management

Week 5

- ✓ Supplier Management
- ✓ Purchase Module
- ✓ Inventory

Week 6

- ✓ Quotation
- ✓ Invoice
- ✓ Payments

Week 7

- ✓ Complaints
- ✓ Service Management
- ✓ Service Tracker

Week 8

- ✓ AMC
- ✓ Installation
- ✓ Employee Management

Week 9

- ✓ Attendance

- ✓ Technician Portal
- ✓ Notifications

Week 10

- ✓ Reports
- ✓ Settings
- ✓ Performance Optimization

Week 11

- ✓ UI Polish
- ✓ Testing
- ✓ Bug Fixes

Week 12

- ✓ Production Deployment
- ✓ Go-Live

Critical Module Dependency Order

